

Fe₅C₂ Nanoparticles: A Facile Bromide-induced Synthesis and as an Active Phase for Fischer-Tropsch Synthesis (FTS)

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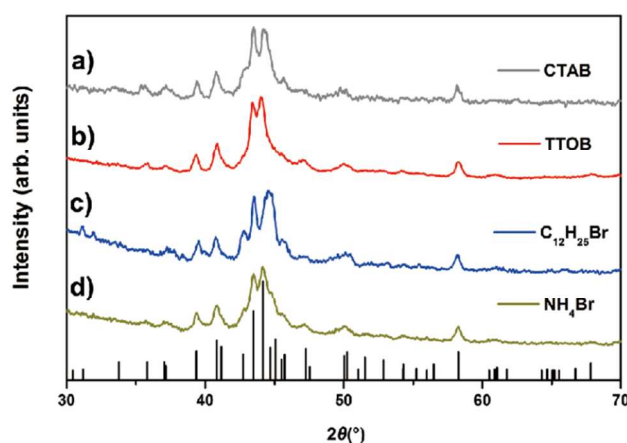


Figure S1. XRD patterns of Fe₅C₂ NPs synthesized from reactions using a) CTAB, b) Tetra-n-octylammonium bromide (TTOB), c) Bromododecane (C₁₂H₂₅Br) and d) Ammonium bromide (NH₄Br) as inducing agents (JCPDS No. 36-1248).

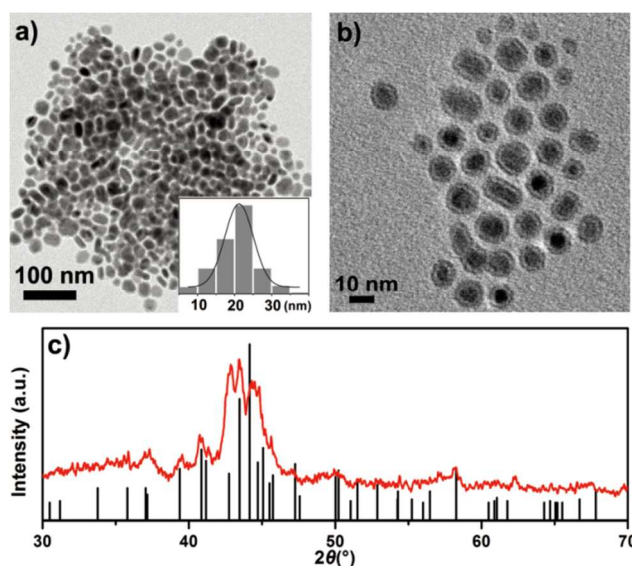


Figure S2. a) TEM image for 20-nm Fe₅C₂ NPs. b) TEM image and c) XRD pattern for 10-nm Fe₅C₂ NPs (JCPDS No. 36-1248). The average particle size estimated from Scherrer's formula was 12.2 nm in diameter

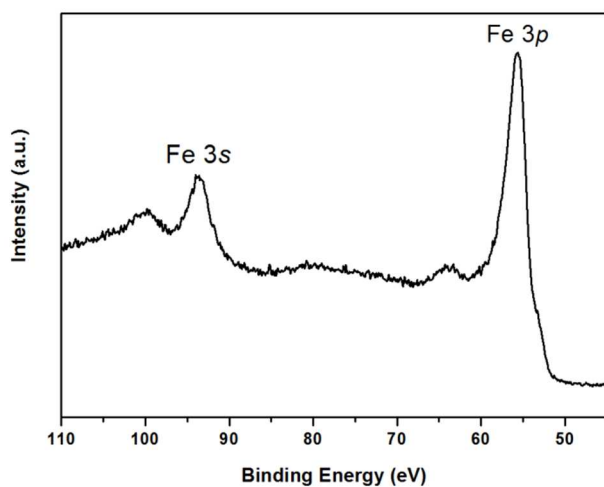


Figure. S3. Fine XPS scan of Fe_5C_2 NPs. The peaks at 93.6 eV and 55.6 eV correspond to the Fe 3s and Fe 3p spectra, respectively.

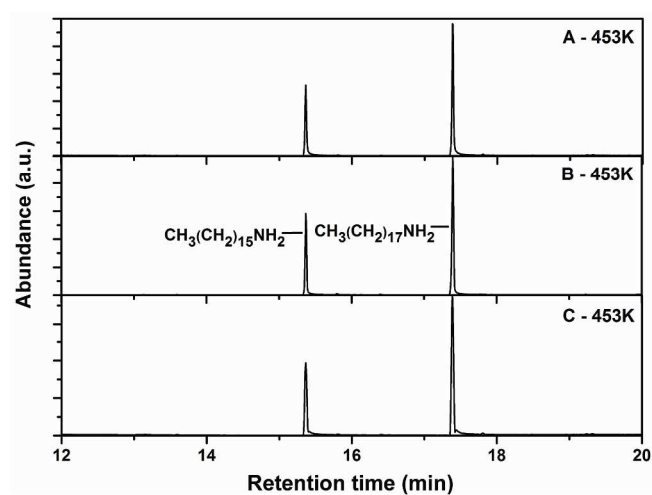


Figure S4. GC-MS chromatograms obtained for reaction solutions extracted at 453 K.

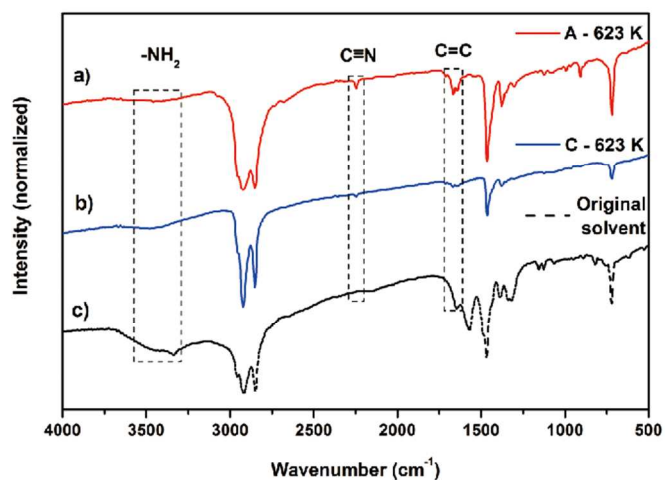


Figure S5. FTIR spectra of a) A-623 K, b) C-623 K and c) the original solvent. The frames indicate the peaks for the corresponding chemical bonds.

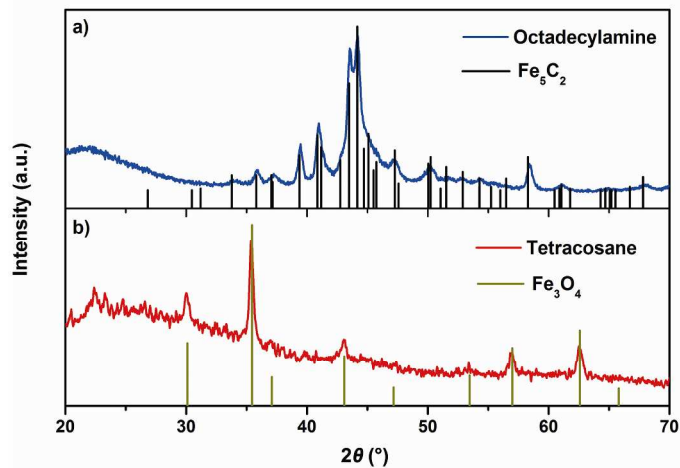


Figure S6. XRD patterns of as-synthesized NPs using a) octadecylamine and b) tetracosane (saturated hydrocarbon) as solvent (Fe_5C_2 : JCPDS No. 36-1248, Fe_3O_4 : JCPDS No. 65-3107). Note that both reactions employed bromide as inducing agent and underwent same process as well.

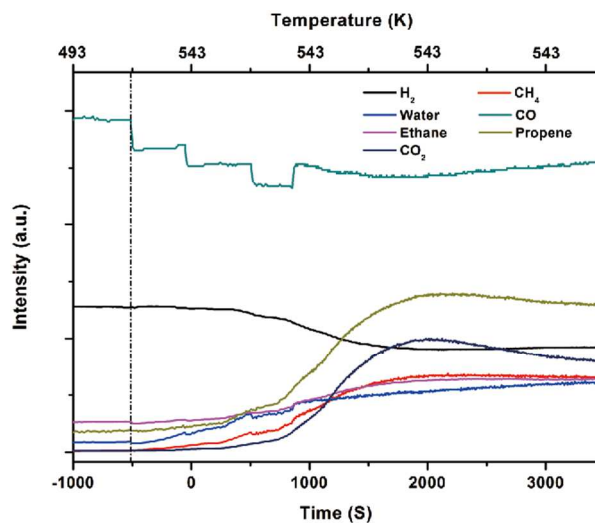


Figure S7. TPSR for reduced Fe_2O_3 catalyst.

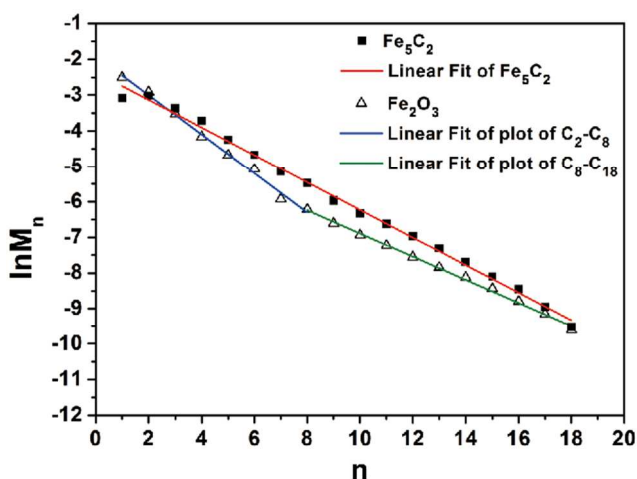


Figure S8. Chain growth probability factor of Fe_5C_2 NPs and reduced Fe_2O_3 . The calculated value of chain growth probability factor α of Fe_5C_2 is 0.68, while that of the reduced iron oxide catalyst presents a dual α value character with 0.55 and 0.72 for C_2 - C_8 products and C_{8+} products, respectively. Note that n is the carbon number of hydrocarbons and M_n is the molar ratio of hydrocarbons with carbon number n (C_n). Chain growth probability factor is generated from the Schulz formula: $\lg(M_n) = (n-1)\lg\alpha + \lg(1-\alpha)$.